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The impact of personality type on online learning performance among high school students

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Abstract
The relationship between individual personality types and academic performance during online learning remains poorly understood. This study analysed extensive sampling data from 4,340 first-year high school students during semesters of face-to-face and online learning, respectively, to investigate the influence of students' Myers−Briggs Type Indicator (MBTI) types on their academic performance in Chinese, mathematics, English, and overall score during online learning. We observed that the online learning environment conferred disadvantages to extroverts, intuitives, thinkers, and perceivers. Specifically, the dimensions of extroversion-introversion and thinking-feeling significantly affected mathematics, while sensing-intuition and judging-perception significantly affected Chinese and English, respectively. Several significant interaction effects between different MBTI dimensions on online learning performance were also found. Girls outperformed boys in online learning performance. However, no sex difference in the effect of personality type on online learning performance was found. Student personality type identification could proactively identify students who may require additional support in the online learning environment and aid in designing effective tools for online education platforms to improve their learning performance. In particular, AI teaching assistants can be integrated into online classrooms as they could help these students address challenges posed by traditional online learning by offering personalised support. These supports include interactive dialogue for extroverts to foster engagement, creative discussion for intuitives to encourage exploration, performance feedback for thinkers to help them adjust their learning strategies, and time-management assistance for perceivers to ensure they stay organised.

Keywords Online learning · Personality · Academic performance · Secondary education · Myers-Briggs personality type indicator (MBTI) · Learning environment

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1 Introduction

Currently, with the development of digitalization and communication technology, online learning is an innovative education pattern. The outbreak of the COVID-19 pandemic accelerated this pattern (Alon et al., 2023). In response to this public health emergency, countries worldwide have taken several measures, including social distancing restrictions and school closures (Chtourou et al., 2020). According to statistics, > 84% of students worldwide have been affected by school closures and left their classrooms during the three-year pandemic (Jackman et al., 2021). Against this backdrop, schools provide students with online learning resources, including recorded lessons and online live-teacher interactions (Camargo et al., 2020). As a flexible approach, online learning has not only provided learning opportunities for students during the pandemic but also demonstrated its necessity during other emergencies (e.g., earthquakes, extreme weather) and long vacations, where traditional face-to-face learning is impractical (Angrist et al., 2022). In addition, many schools in rural areas lack qualified teachers, and online learning can provide students in these areas with access to high-quality urban or international educational resources (Stenman & Pettersson, 2020). Online learning also makes it possible for people with disabilities to be educated at home, promoting equity in education (Meleo-Erwin et al., 2021). Online learning has gradually become an indispensable part of the education system, and its impact is profoundly shaping the face of future education.

Although online learning has brought numerous possibilities to future education, some people remain hesitant to adopt this technology in the classroom (Al-Obaydi, 2023). This hesitation may arise from the belief that online learning systems are only suitable for certain students and may not meet the needs of all students (Al-Obaydi, 2020). To address this issue, Rodríguez-Ardura and Meseguer-Artola (2016) proposed that more humanistic learning support should be incorporated to enhance students' online learning experience. The main ideas of humanistic learning, as developed by Carl Rogers, Abraham Maslow, and Combs, emphasize students' self-actualization and the importance of fostering their personal development (Chen & Schmidtke, 2017; Al-Obaydi, 2023). According to humanistic learning theory, personality traits can significantly influence key aspects of learning, such as attitudes and behaviours (Al-Obaydi, 2023). This understanding highlights that individuals with different personality types may perform differently in online learning environments. Recognizing these differences can help identify students who may require additional support in online learning and provide insights for developing new teaching strategies to improve their learning experience and performance.

Nevertheless, there is a lack of research on how students with different personality perform in online learning, especially for different subjects. Hence, this study aims to employ large-scale empirical data to investigate the influence of students' personality types on their online learning performance.

1.1 Online learning performance

The impact of online learning on students' academic performance remains a subject of debate. Some studies have suggested that online learning is not as effective as face-to-face learning. Engzell et al. (2021) evaluated the impact of school

closures on primary school students in the Netherlands using a substantial dataset ($N \approx 350{,}000$) and found that the academic loss was approximately 3% points, equivalent to one-fifth of a school year (i.e., 8 weeks) of learning. Evidence from the Italy (Contini et al., 2022), Germany (Schult et al., 2022), the United States (Halloran et al., 2021), and Brazil (Lichand et al., 2022) also showed that teaching progress during the pandemic period was slow and that students' academic performance declined. The reasons led to academic performance loss included the limitations posed by the inability to access high-quality digital learning devices and stable internet connections (Domina et al., 2021), the lack of supervision and timely interactions between teachers and students (Wong, 2023; Yu, 2021).

Some studies continued to explore the lasting impact of school closures and argued that it is inaccurate to generalize the initial experiences of academic losses due to online learning into a longer-term trend, as schools and students will adapt to the online environment over time (Birkelund & Karlson, 2023; Meeter, 2021). Research from Belgium (Gambi & De Witte, 2021), Brazil (Lichand et al., 2022), and the Netherlands (Haelermans et al., 2021) found that students did not experience further academic losses during the subsequent school closures. Additionally, students may even improve their learning performance due to more focused learning at home or increased family tutoring (Velde et al., 2021). Velde et al. (2021) used language learning data from over 133,000 middle school students in the Netherlands and found that students' learning progress was not negatively affected by the shift to online learning, and the accuracy and response time of open-ended questions improved. Similarly, a study from Germany reported that students' mathematical performance improved during the period of online learning compared to the previous year (Spitzer & Musslick, 2021).

In fact, the effectiveness of online learning cannot be generalized. In other words, students' participation and performance in online learning are influenced by a variety of factors. Students' attributes (e.g., age, sex, socioeconomic background) have been widely studied to explain differences in academic performance. Recent research has shown that younger students (Tomasik et al., 2021), girls (Lichand et al., 2022) and students in low-income communities (Hill & Reimer, 2024) are more likely to be adversely affected by online learning. Furthermore, evidence indicates that variations in academic performance during online learning differ across subjects (Gambi & De Witte, 2021; Kuhfeld et al., 2020; Zierer, 2021). For instance, Gambi and De Witte (2021) found that one year after the COVID-19 school closures, students' science score caught up with the previous cohorts, social science significantly surpassed the previous cohorts, while mathematics, Dutch (mother tongue), and French (foreign language) continued to suffer. However, existing studies doesn't take the influence of personality on online learning performance into consideration.

1.2 The influence of personality type

As a stable individual attribute, personality has always been considered an important factor in determining students' learning motivation, learning process and academic performance (De Raad & Schouwenburg, 1996; Komarraju et al., 2011), with several studies suggesting that personality can influence students' perceptions of and

satisfaction with online learning (Bhagat et al., 2019; Bolliger & Erichsen, 2013; Harrington & Loffredo, 2010). Nevertheless, there is a lack of research on how students' personalities affect their academic performance in online learning, especially for different subjects. This study aims to investigate the influence of students' personality types on their online learning performance.

We used the well-established Myers−Briggs Personality Type Indicator (MBTI) to measure students' personality types because this model has been extensively validated (Choi, 2021) and widely used in the education field (Kang, 2020). The MBTI model was developed by Myers and Briggs (Myers et al., 1998) and was based on Jung's theory of psychological types (Pittenger, 1993). This model consists of four dimensions with opposite poles: extroversion-introversion (EI), sensing-intuition (SN), thinking-feeling (TF), and judging-perception (JP) (Furnham, 2022). Individuals have one pole superior to the other on each of these dimensions, and their preferences on each dimension are independent of those on the other three dimensions (Offir et al., 2007). Therefore, the combination of the four dimensions produces sixteen personality types (i.e., 16 personalities) (Myers et al., 1998).

The EI dimension reflects the way individuals acquire energy. Extroverts, who draw energy from interacting with the external world and enjoy socializing and active occasions, tend to respond more positively than introverts when faced with group learning and discussion (Dewar & Whittington, 2000). Introverts, on the other hand, replenish their energy from their inner world and have a natural inclination for solitude, and thus prefer independent and self-directed learning (Bhagat et al., 2019).

The SN dimension reflects the way individuals obtain information. Sensors predominantly depend on their own senses to observe occurrences and pay more attention to objective facts, while intuitives often gravitate towards abstract discussions over concrete topics and tend to focus more on imagination, unreachable meaning, and possibilities. Sensors perform significantly better than intuitives in practical courses, while the opposite is observed in courses with higher levels of abstract content (M. Ayoubi & Ustwani, 2014).

The TF dimension reflects an individual's tendency to make decisions. Thinkers primarily utilize logic for decisions and evaluations, with great emphasis on structure and principles, while feelers base their decisions more on emotions than on logic and are better at capturing and understanding other's emotions. In courses that emphasize logic and objective data, thinkers excel over feelers, and feelers are more likely to withdraw from the courses even if they have good academic performance (Felder & Brent, 2005).

The JP dimension reflects the way individuals organise their lives and handle the external world. Judgers have a penchant for organising the external world according to plans and tend to control situations while perceivers favour a flexible and spontaneous lifestyle. As for learning, judgers perform better when faced with heavy course loads and corresponding time management demands (Felder & Brent, 2005), while perceivers are better adapted to flexible learning tasks (DiRienzo et al., 2010).

Some studies implied that students tend to prefer learning pattern that match their personality types. When comparing online learning and face-to-face learning, Harrington and Loffredo (2010) distributed a questionnaire to 166 college students and found introverts had a preference for online courses, while extroverts preferred

face-to-face courses. This may be because extroverts receive more external rewards, such as praise and feedback, through interactions with classmates and teachers in face-to-face settings. According to behaviorist learning theory, these external stimuli reinforce extroverts' learning motivation and shape their behaviour (Muhajirah, 2020), making them more inclined to choose this environment instead of online learning. Bolliger and Erichsen (2013) conducted an online satisfaction questionnaire with 72 students enrolled in online courses at two universities. Their results indicated that sensing online learners were more satisfied with this learning pattern of using chat tools compared to intuitives. From the constructivist learning theory perspective, a possible reason for this is that sensors who focus on concrete information, find it easier to construct knowledge through the materials and videos provided in online learning environments (Muhajirah, 2020). However, all of these studies focused on the influence of personality type on subjective perceptions and motivation in online learning. Despite the consensus that the perceptions of learning can affect behaviour (Irani et al., 2000), there is a lack of research on the impact of personality type on learning performance in online learning environments.

2 Methods

2.1 Participants

This study was conducted in Hainan Province, China. Including only one province can minimize the influence of different economic development levels, epidemic prevention and control measures, and exam requirements between different provinces on students' academic performance. Seven level-1 high schools in Hainan Province (Appendix Table A.1), representing approximately 30.43% of all level-1 high schools in Hainan Province (a total of 23 schools), were selected in this study. School grades, recognized by the Education Department of Hainan Province, are determined as a result of a comprehensive assessment of the school's educational conditions and resources. Compared to other level schools, level-1 high schools have better facilities and faculty teams. Therefore, including only level-1 high schools can help reduce the impact of different teaching qualities, faculty, and facilities on students' academic performance.

Our data included 4,340 first-year high school students with an average age of 16.32 years. The sample comprised 1,978 boys and 2,362 girls. The proportion of female students in our sample was 54.42%, which is similar to the average percentage of female students in Chinese high schools (50.71%) (Ministry of Education of the People's Republic of China, 2020).

2.2 Procedure

The COVID-19 pandemic provided a unique natural experiment for online learning research. Before the pandemic, all schools in Hainan Province conducted face-to-face learning in the first semester of the 2019–2020 academic year (i.e., from September 2019 to January 2020). In this semester, invitations to participate in the MBTI test were

extended to all 4,691 first-year high school students from these seven schools, 4,524 of whom completed the questionnaires in class. In early January 2020, students took their first-semester final exams. Subsequently, the COVID-19 pandemic emerged, and from February onwards, all first-year high school students from the seven schools underwent online learning for the entire second semester, culminating in online final exams in July. A total of 133 participants without exam results (e.g., due to illness-related absences) were excluded from the study, and 51 participants with problems in their original MBTI responses (e.g., incomplete responses, consistent response pattern) were also excluded. Ultimately, 4,340 valid participants were included in the study.

2.3 Measurement

2.3.1 Personality types

The MBTI questionnaire for high school students (Appendix Section B) measured the participants' personality types. This questionnaire was developed based on traditional MBTI (Myers et al., 1998) and was tailored to the characteristics of Chinese high school students (He, 2016). During the questionnaire revision process, 4,554 first-year high school students from ten cities in Zhejiang Province, China, were randomly selected to participate in a three-phase process. The first revision was based on a preliminary test, while the second and third revisions were based on large-scale tests conducted after in-depth interviews with students and psychology experts. The final version of the questionnaire was proven to have good reliability (i.e., internal consistency, split-half reliability, and test-retest reliability) and validity (i.e., construct validity, content validity, and criterion-related validity) and was approved by an expert review (Cao et al., 2016). Specifically, the criterion-related validity analysis employed by the MBTI developed by Cai et al. (2001) is widely used in China as the criterion (Zhu & Xie, 2010). The obtained results confirmed our questionnaire's excellent criterion validity.

Our questionnaire consisted of 80 multiple-choice questions, with 19 questions for each of the four dimensions (EI, SN, TF, and JP), and four questions served as content validity indicators. When answering questions, participants had to choose one of the dichotomous answers under a short sentence description to reveal their preferences in this dimension. According to the scores obtained by the participants on the four dimensions, they were classified into one of 16 personality groups. The Cronbach's α coefficients for the four dimensions of EI, SN, TF, and JP were 0.84, 0.72, 0.77, and 0.85, respectively, indicating a high level of internal consistency in this test. The correlation coefficients between the total scores of each dimension and the content validity indicator of that dimension were 0.576 ($p < 0.001$), 0.332 ($p < 0.001$), 0.528 ($p < 0.001$), and 0.634 ($p < 0.001$), indicating a moderate to strong level of content validity.

2.3.2 Academic performance

Students' final exam performance was used to reflect their academic performance for the semester, as their emphasis on final exams allowed them to fully demonstrate their learning abilities. On the other hand, schools also used final exams to comprehensively

evaluate students' learning and mastery of each subject. Since the final exam papers were independently set by each school, with varying test-setting strategies and routines, this could lead to inconsistencies in the difficulty level of the exams across schools. Therefore, using scores to measure final exam performance would be inappropriate. Additionally, to avoid fluctuations in scores due to differences in course content and difficulty across semesters, we used the students' ranking proportion (calculated as exam ranking within one's school divided by the total number of participants in the school, hereinafter referred to as 'RP(s)' in the following) to represent students' academic performance. The smaller the RP value is, the better the academic performance of the student. Given that online learning performance may vary across different subjects, we obtained RPs for students in three subjects—Chinese, mathematics, and English—which are generally considered core subjects in China. Chinese represents the mother tongue and is a key subject for assessing language comprehension and analytical skills (Zheng, 2022), while mathematics reflects logical reasoning and problem-solving capabilities (Hoyles, 2018). English, as a foreign language, is essential for language acquisition and communication skills (Ma et al., 2024). These three subjects are compulsory in the Nationwide Unified Examination for Admissions to General Universities and Colleges across China. The RPs for the overall score of these three subjects were also calculated and analysed to reflect the overall performance.

Furthermore, we assessed the impacts of personality type on online learning performance by examining students' variation in the RPs between the two semesters (calculated as the RP in the second semester final exam minus the RP in the first semester final exam). A negative variation in the RPs indicates that the student's academic performance has improved. A positive variation in the RPs indicates that the student's academic performance has declined.

2.4 Data analysis

Statistical analysis was performed using R software. Chi-square test was used to examine whether there is a significant difference in the proportions of personality types between the sexes. Means and 95%CI were calculated to display the descriptive statistics of RPs and variations in RPs across all subjects. Data were assessed for normality with the Shapiro–Wilk test. Spearman correlation analysis was used to examine the relationship between the variation in RPs across different subjects and personality types in the four dimensions. Spearman correlation analysis was suitable for this study because the personality type data didn't follow the normal distribution. Multifactor analysis of variance (MANOVA), with the four personality dimensions as factors, was used to test the effect of each dimension on variation in RPs across different subjects. MANOVA was chosen because it allows for the testing of multiple independent variables (i.e., four personality dimensions), as well as the interaction effects between different dimensions. Two-tailed t-tests were used for further analysis of the interaction effects, as they were appropriate for examining whether there were significant differences between two sample means without assuming the direction of the effect. MANOVA was also used to estimate the homogeneity in the results by incorporating sex as a new factor. Statistical significance was set at $p < 0.05$.

3 Results

3.1 Demographics

A total of 4,340 first-year high school students participated in the study. Most students reported that they were extroverts, sensors, feelers, and perceivers, and this pattern was observed in both boys and girls (Fig. 1). The results of the chi-square test showed that across all four dimensions, there were no differences in the proportions of personality types between the sexes (Appendix Table A.2). Moreover, all 16 personality traits were represented in the sample. The distributions of 16 personalities among students, categorized by sex, are detailed in Appendix Table A.3. ESTJ students were the most common, followed by ISFP students. Among boys, ESTJ was the most common, while among girls, ISFP was the most common.

3.2 Descriptive statistics

The descriptive statistics for the RPs in Chinese, mathematics, English, and the overall score on the two final exams of the students across the four dimensions are presented in Table 1. The variation in RPs across the four dimensions is shown in Table 2. The results of the Shapiro−Wilk test indicated that the variation in RPs in these three subjects and the overall score between the two final exams were normally distributed ($p's < 0.001$). Table 2 suggests that introverts, sensors, feelers, and judgers improved their academic performance as their overall score RPs decreased. In contrast, extroverts, intuitives, thinkers, and perceivers performed worse in learning, as their variation in overall score RPs was positive.

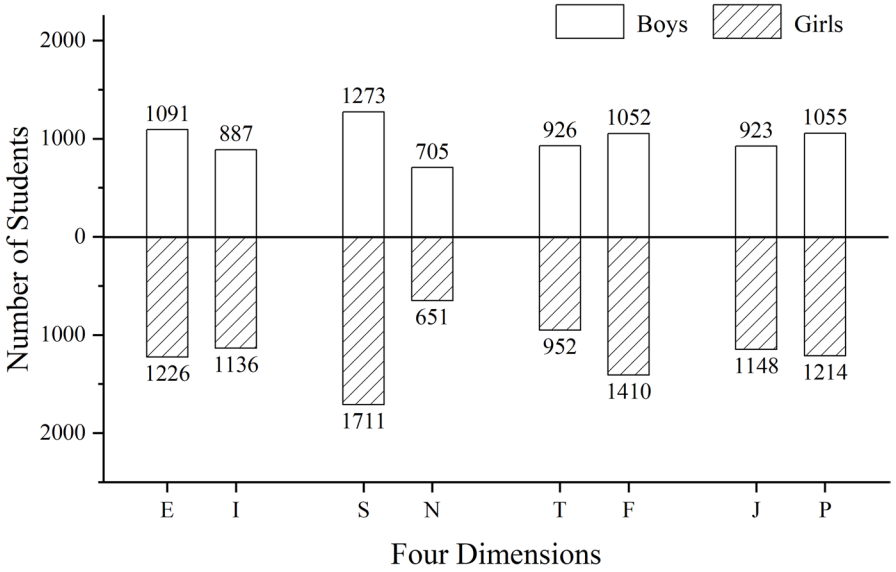

Fig. 1 Personality types in the four dimensions of the students. The hollow and striped bars represent the number of boys and girls, respectively ($N = 4{,}340$)

3.3 Intercorrelations

Spearman correlation analysis was carried out using the variation in RPs and personality types in the four dimensions as variables (Table 3). The correlations between the variation in RPs in overall score (i.e., R4) and each of the three subjects (i.e., R1, R2, and R3) proved to be statistically significant (R1: $\rho=0.437$, $p<0.001$; R2: $\rho=0.677$, $p<0.001$; R3: $\rho=0.548$, $p<0.001$), indicating that performance in an individual subject contributed to overall performance, but it did not entirely determine overall performance. The correlations between all four dimensions and the variation in overall score RPs were significant (EI: $\rho=0.033$, $p=0.028$; SN: ρ = -0.033, $p=0.031$; TF: $\rho=0.038$, $p=0.012$; JP: ρ = -0.030, $p=0.047$). Regarding the correlations between the four dimensions and specific courses, EI was positively correlated with the variation in RPs in mathematics ($\rho=0.039$, $p=0.009$), SN was negatively correlated with the variation in RPs in Chinese (ρ = -0.029, $p=0.050$), TF was positively correlated with the variation in RPs in mathematics ($\rho=0.036$, $p=0.018$), and JP was negatively correlated with the variation in RPs in English (ρ = -0.031, $p=0.044$).

3.4 The impacts of personality type

MANOVA was then conducted to explore whether there were significant differences in the variation in RPs among students with different personality types (Table 2).

In the EI dimension, significant differences were observed between extroverts and introverts in the variation in mathematics (extroverts: $M=0.009$, 95% CI=0.001 to 0.018; introverts: $M=-0.007$, 95% CI = − 0.016 to 0.002; $F_{1,4324}=6.761$, $p=0.009$) and overall score RPs (extroverts: $M=0.006$, 95% CI = − 0.001 to 0.012; introverts: $M=-0.005$, 95% CI = − 0.013 to 0.002; $F_{1,4324}=4.840$, $p=0.028$). There was a significant interaction effect between the EI dimension and the TF dimension in terms of the variation in mathematics RPs ($F_{1,4324}=4.731$, $p=0.030$). For students who were thinkers, IT significantly outperformed ET (IT: $M=-0.007$, 95% CI = − 0.021 to 0.007; ET: $M=0.024$, 95% CI=0.012 to 0.037; $t=3.280$, $p=0.001$), while for feelers, there was no significant difference between IF and EF (IF: M = -0.007, 95% CI = − 0.019 to 0.005; EF: $M=-0.003$, 95% CI = − 0.014 to 0.009; $t=0.543$, $p=0.587$).

In terms of the SN dimension, significant differences were observed between sensors and intuitives in the variation in Chinese (sensors: $M=-0.005$, 95% CI = − 0.015 to 0.005; intuitives: $M=0.012$, 95% CI = − 0.002 to 0.027; $F_{1,4324}=3.864$, $p=0.049$) and overall score RPs (sensors: $M=-0.003$, 95% CI = − 0.009 to 0.003; intuitives: $M=0.009$, 95% CI = − 0.001 to 0.018; $F_{1,4324}=3.925$, $p=0.048$).

For the TF dimension, significant differences were observed between thinkers and feelers in the variation in mathematics (thinkers: $M=0.010$, 95% CI=0.001 to 0.020; feelers: $M=-0.005$, 95% CI = − 0.013 to 0.003; $F_{1,4324}=5.591$, $p=0.018$) and overall score RPs (thinkers: $M=0.008$, 95% CI=0.000 to 0.015; feelers: M = -0.005, 95% CI = − 0.012 to 0.002; $F_{1,4324}=6.436$, $p=0.011$). There was a significant interaction effect between the TF dimension and the JP dimension on the variation in overall score RPs ($F_{1,4324}=4.672$, $p=0.031$). For students characterized as judgers, FJ significantly outperformed TJ (FJ: $M=-0.018$, 95% CI = − 0.029 to − 0.008; TJ: $M=0.007$, 95%

Table 1 RPs in three subjects and overall score in the two final exams

		the 1st semester		the 2nd semester	
		Mean	95% CI	Mean	95% CI
Chinese					
EI	E	0.509	(0.497, 0.520)	0.507	(0.495, 0.519)
	I	0.464	(0.451, 0.476)	0.466	(0.454, 0.479)
SN	S	0.489	(0.479, 0.500)	0.485	(0.474, 0.495)
	N	0.484	(0.468, 0.499)	0.496	(0.481, 0.511)
TF	T	0.493	(0.480, 0.506)	0.496	(0.482, 0.509)
	F	0.483	(0.472, 0.495)	0.482	(0.471, 0.494)
JP	J	0.487	(0.475, 0.499)	0.480	(0.468, 0.493)
	P	0.488	(0.476, 0.500)	0.495	(0.484, 0.507)
Mathematics					
EI	E	0.511	(0.499, 0.523)	0.520	(0.509, 0.532)
	I	0.473	(0.461, 0.486)	0.466	(0.454, 0.479)
SN	S	0.495	(0.485, 0.505)	0.494	(0.483, 0.504)
	N	0.490	(0.474, 0.505)	0.499	(0.483, 0.514)
TF	T	0.461	(0.448, 0.474)	0.471	(0.458, 0.484)
	F	0.518	(0.507, 0.530)	0.514	(0.502, 0.525)
JP	J	0.484	(0.472, 0.496)	0.484	(0.472, 0.497)
	P	0.502	(0.490, 0.514)	0.505	(0.493, 0.517)
English					
EI	E	0.511	(0.499, 0.523)	0.514	(0.502, 0.525)
	I	0.477	(0.464, 0.489)	0.476	(0.463, 0.488)
SN	S	0.493	(0.482, 0.503)	0.492	(0.482, 0.503)
	N	0.500	(0.485, 0.516)	0.503	(0.488, 0.519)
TF	T	0.501	(0.487, 0.514)	0.506	(0.493, 0.519)
	F	0.491	(0.479, 0.502)	0.488	(0.477, 0.499)
JP	J	0.486	(0.474, 0.499)	0.482	(0.469, 0.494)
	P	0.503	(0.491, 0.515)	0.509	(0.497, 0.521)
Overall score					
EI	E	0.520	(0.508, 0.531)	0.525	(0.513, 0.537)
	I	0.473	(0.460, 0.486)	0.468	(0.455, 0.480)
SN	S	0.498	(0.487, 0.508)	0.495	(0.484, 0.505)
	N	0.498	(0.483, 0.514)	0.507	(0.491, 0.522)
TF	T	0.485	(0.471, 0.498)	0.492	(0.479, 0.506)
	F	0.508	(0.497, 0.519)	0.503	(0.492, 0.514)
JP	J	0.488	(0.476, 0.500)	0.483	(0.471, 0.496)
	P	0.507	(0.495, 0.519)	0.512	(0.500, 0.524)

CI = − 0.003 to 0.017; t=3.413, p=0.001), while for perceivers, there was no significant difference between FP and TP (FP: M=0.003, 95% CI = − 0.005 to 0.012; TP: M=0.009, 95% CI = − 0.002 to 0.021; t=0.839, p=0.402).

In the JP dimension, there was a significant interaction effect between the JP dimension and SN dimension on the variation in Chinese RPs ($F_{1,4324}$ = 5.101, p=0.024).

Table 2 Variation in RPs in three subjects and overall score between the two final exams

		Chinese				Mathematics				English				Overall score			
		Mean	95% CI	F	*p*	Mean	95% CI	F	*p*	Mean	95% CI	F	*p*	Mean	95% CI	F	*p*
EI	E	−0.001	(−0.012, 0.010)	0.216	0.642	0.009	(0.001, 0.018)	6.761	**0.009**	0.003	(−0.004, 0.010)	0.502	0.479	0.006	−0.001, 0.012)	4.840	**0.028**
	I	0.003	(−0.009, 0.015)			−0.007	(−0.016, 0.002)			−0.001	(−0.008, 0.006)			−0.005	(−0.013, 0.002)		
SN	S	−0.005	(−0.015, 0.005)	3.864	**0.049**	−0.002	(−0.009, 0.006)	1.777	0.183	0.000	(−0.006, 0.006)	0.312	0.576	−0.003	(−0.009, 0.003)	3.925	**0.048**
	N	0.012	(−0.002, 0.027)			0.009	(−0.002, 0.020)			0.003	(−0.006, 0.012)			0.009	(−0.001, 0.018)		
TF	T	0.002	(−0.010, 0.015)	0.234	0.628	0.010	(0.001, 0.020)	5.591	**0.018**	0.005	(−0.002, 0.013)	2.420	0.120	0.008	(0.000, 0.015)	6.436	**0.011**
	F	−0.001	(−0.012, 0.010)			−0.005	(−0.013, 0.003)			−0.002	(−0.009, 0.004)			−0.005	(−0.012, 0.002)		
JP	J	−0.007	(−0.019, 0.005)	1.928	0.165	0.000	(−0.009, 0.009)	0.737	0.390	−0.004	(−0.012, 0.003)	5.653	**0.018**	−0.005	(−0.012, 0.002)	5.451	**0.020**
	P	0.007	(−0.004，0.018)			0.003	(−0.006，0.012)			0.006	(−0.001, 0.013)			0.005	(−0.001, 0.012)		
Interaction		SN x JP		5.101	**0.024**	EI x TF		4.731	**0.030**					TF x JP		4.672	**0.031**

Table 3 Intercorrelations for the variables

	R1	R2	R3	R4	P1	P2	P3	P4
R1 the variation in Chinese RPs	1.000							
R2 the variation in mathematics RPs	0.011	1.000						
R3 the variation in English RPs	0.036*	0.069***	1.000					
R4 the variation in overall score RPs	0.437***	0.677***	0.548***	1.000				
P1 EI	−0.007	0.039**	0.011	0.033*	1.000			
P2 SN	−0.029*	−0.023	−0.009	−0.033*	−0.084***	1.000		
P3 TF	0.006	0.036*	0.023	0.038*	0.021	0.032*	1.000	
P4 JP	−0.026	−0.007	−0.031*	−0.030*	0.063***	0.216***	0.218***	1.000

**** Indicates a correlation that is significant at the 0.001 level, ** indicates a correlation significant at the 0.01 level, and * indicates a correlation significant at the 0.05 level*

For students who were intuitives, NJ significantly outperformed NP (NJ: $M = -0.015$, 95% CI = − 0.042 to 0.013; NP: $M=0.025$, 95% CI=0.008 to 0.042; $t = -2.408$, $p=0.016$), while for sensors, there was no significant difference between SJ and SP (SJ: $M = -0.005$, 95% CI = − 0.018 to 0.009; SP: $M = -0.005$, 95% CI = -0.019 to 0.009; $t=0.036$, $p=0.971$). In addition, significant differences were observed between judgers and perceivers in the variation in English (judgers: M = -0.004, 95% CI = − 0.012 to 0.003; perceivers: $M=0.006$, 95% CI = − 0.001 to 0.013; $F_{1,4324} = 5.653$, $p=0.018$) and overall score RPs (judgers: $M = -0.005$, 95% CI = − 0.012 to 0.002; perceivers: $M=0.005$, 95% CI = − 0.001 to 0.012; $F_{1,4324} = 5.451$, $p=0.020$).

3.5 Homogeneity across different sexes

We proceeded with MANOVA to estimate the homogeneity in the results by incorporating sex as a new independent variable. No interaction effects between sex and any of the personality types in the four dimensions on the variation in RPs were observed (Table 4), indicating that the impacts of personality type on online learning performance did not exhibit heterogeneity across sexes. Figure 2 displays the homogeneous impacts of personality type on the variation in RPs by sex. Interestingly, the variation in RPs across all subjects and personality types (except Chinese for sensors) was greater for boys than for girls, indicating that girls may outperform boys in online learning performance.

4 Discussion

4.1 Theoretical implications

In existing research on intervention methods for online learning, students are generally considered a homogeneous whole, where a one-size-fits-all approach is applied to address the needs of all students in improving their academic performance. Our study elucidated differences in learning performance among students with different personality types regarding online learning and identified students who may encounter obstacles. The findings can enable teachers to proactively pay more attention to and support these students. Furthermore, the results revealed that the characteristics of a subject determine the direction and extent of the impacts of personality type on online learning performance.

In terms of Chinese, we found that online learning is less beneficial for intuitives than for sensors. This may be because Chinese is a student's mother tongue, and its study often goes beyond mastering grammar and vocabulary. It typically involves a deep understanding of the language and literary analysis. Intuitive students learn through insight and intuition and often enjoy abstract thinking and innovation. Face-to-face classes provide more opportunities for open discussions and free exploration (Kalogeropoulos et al., 2021), which caters to their preferences for creative thinking. In contrast, online learning may focus more on standardized teaching content and structured learning processes (Offir et al., 2007), which limits

Table 4 Interaction effects between sex and personality types in four dimensions

	Chinese		Mathematics		English		Overall score	
	F	*p*	*F*	*p*	*F*	*p*	*F*	*p*
sex x EI	0.525	0.469	0.952	0.329	0.107	0.743	0.002	0.961
sex x SN	3.724	0.054	0.422	0.516	0.558	0.455	1.738	0.187
sex x TF	0.544	0.461	0.220	0.639	0.093	0.760	0.583	0.445
sex x JP	0.308	0.579	0.717	0.397	0.060	0.806	0.017	0.897

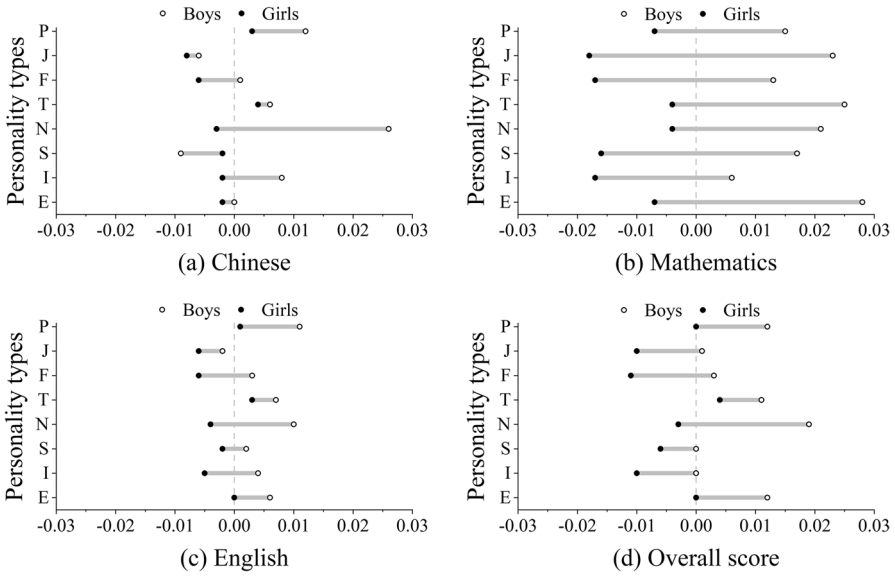

Fig. 2 Homogeneous impacts of personality type on the variation in RPs by sex. The white and black dots indicate the variation in RPs between the two final exams for boys and girls, respectively, with specific personality types. The grey bars represent the difference between the sexes. The variation in RPs in Chinese (**a**), mathematics (**b**), English (**c**), and the overall score (**d**) are presented in four separate subfigures. Data from $N = 1{,}978$ boys and 2,362 girls, totalling 4,340 first-year high school students from Hainan Province, China, were used in the analyses

the space for free exploration and therefore may lead to a decreased understanding of the language among intuitives. As a result, their academic performance in Chinese may decline. In addition, intuitives are less adept than sensors at absorbing knowledge by watching video lectures or reading online materials (Irani et al., 2000), which might also contribute to their underperformance in such learning environments. This result extends the findings of Bolliger and Erichsen (2013), which indicated that sensing learners were more satisfied with online learning than intuitives, as it explores the impact of personality types on learning performance in online environments.

Regarding mathematics learning, it has been observed that extroverts are more likely to face disadvantages than introverts in online learning environments. Mathematics education requires students to first absorb and understand concepts, formulas and problem-solving methods imparted by teachers before applying this knowledge to solve problems (Hoyles, 2018). During face-to-face learning, extroverts who struggle to understand the material have the advantage of direct communication with people around them to seek clarification (Komarraju et al., 2011). However, when learning online, the real-time verbal interaction between students and teachers is reduced (Yu, 2021; Yuan & Kim, 2014), potentially delaying the resolution of their queries. In contrast, the common interaction method provided by online platforms is to ask teachers questions or provide feedback through text comments (Miller, 1993), which aligns well with introverts' habits and makes them feel less pressured (Deborah et al., 2015). Therefore, they feel comfortable and are more involved in class. Additionally, solving mathematics problems often demands great logical ability and thinking space for calculations and reasoning. Online learning provides introverted students with their own space, which is quieter and less disruptive than the campus environment (Bolliger & Erichsen, 2013; Harrington & Loffredo, 2010). They could study at their own pace, concentrate and think carefully (Bhagat et al., 2019). Hence, they achieve good academic performance. This result is consistent with the findings of Harrington and Loffredo (2010), which showed that introverts preferred online courses compared to extroverts. While their research focused solely on learner preferences, our study goes a step further.

We also found that compared with feelers, thinkers tend to perform worse in online mathematics learning. As previously mentioned, the knowledge taught in mathematics classes is often interconnected and sequential. Failure to master earlier content may hinder understanding of subsequent content. Hence, maintaining pace with the teacher is critical for students' study of mathematics (Attard & Holmes, 2022). Thinkers, who are typically logic-driven and value objective standards and evaluations (Rovai & Grooms, 2004), require feedback on their performance in their learning process to make appropriate adjustments. In face-to-face learning environments, teachers and peers serve as crucial sources of feedback and information (Bostic et al., 2021; Huang et al., 2023). They provide objective assessments that enable thinkers to adjust their understanding and keep pace with the progression of teaching. However, real-time interaction and feedback between teachers and students are lacking in online learning environments (Wong, 2023), which makes it challenging for thinkers to quickly identify where they need to adjust. Consequently, it impacts their learning efficiency and outcomes. Conversely, feelers, who primarily base their decisions on subjective emotions rather than objective criteria, seem less affected by the shift to online learning.

In terms of English, our findings indicate that perceivers were at a disadvantage compared to judgers when learning online. English is a student's foreign language, and thus, its learning primarily focuses on mastering grammar and vocabulary. This necessitates dedicating time outside class to assignment and review sessions to memorize and practice these language elements for effective usage. During face-to-face learning, students typically perform assignments and review knowledge under the watchful eyes of teachers during designated self-study periods. However, the

online learning environment, which lacks direct supervision (Sullivan et al., 2020), requires students to plan for learning well and be highly self-disciplined (Bai & Gu, 2024). Judgers inherently value structure and possess strong time-management skills, enabling them to excel in organising and completing assignments (DiRienzo et al., 2010; Swope & Schmitt, 2006), as well as regular review and exam preparation. They are more likely to maintain learning behaviours and thus are more likely to achieve better academic performance. Conversely, perceivers, who often prefer flexibility over structure and tend to have more relaxed time-management skills, might find it challenging to organise and complete assignments, engage in regular review, and prepare for exams as efficiently as judgers. Although some studies suggest that the freedom and flexibility offered by online learning are appealing to perceivers (Rovai & Grooms, 2004), our result indicates that this very flexibility can become a challenge for them.

In our research, the results of interactions indicate that significant differences in academic performance that exist in a single MBTI dimension may not be significant when combined with other dimensions. For instance, the single EI dimension impacted online learning performance in mathematics. However, when combining the EI and TF dimensions, it was found that for students with a feeling preference, there was no difference between introverts (IF) and extroverts (EF). On the other hand, dimensions that did not initially show significant differences may become significant when combined with others. For example, the single JP dimension did not impact online learning performance in Chinese. However, for students with an intuitive preference, judgers (NJ) performed better than perceivers (NP). This suggests that predicting student academic performance may require not only the consideration of a single MBTI dimension but also an examination of interactions with other dimensions.

Furthermore, the girls derived advantages from online learning pattern, whereas the opposite was true for boys. However, a previous study conducted in Brazil suggests that girls suffer more losses in online learning than boys (Lichand et al., 2022). A study in Bangladesh also reveals that during the pandemic, girls exhibited a more pronounced decline in their motivation to learn and their desire for education, which may be due to gender norms and societal expectations regarding paid work and household responsibilities (Seager et al., 2022). These inconsistent results suggest that when applying research findings on factors affecting online learning performance, the influence of societal and cultural backgrounds should be considered.

4.2 Practical implications

The above analyses revealed that students who rely on supervision, communication, and feedback may lose motivation and become ineffective due to the lack of interpersonal interaction in online learning. To increase student engagement, we suggest incorporating additional online classroom tools. First, using an AI teaching assistant (AITA) may offer a possible solution for engaging students in learning from home in a more interactive classroom environment (Kim et al., 2020). By integrating technologies such as voice and facial recognition, information extraction, and

text analysis (Chen et al., 2023), AITAs can provide timely, one-on-one classroom observation, feedback, response and supervision services to students (Chiu et al., 2023; Taneja et al., 2024; Vacalopoulou et al., 2024). Through these functions, the AITA can help students overcome some of the challenges posed by traditional online learning, optimise their online learning experience, and achieve better academic performance. For instance, when an extrovert asks questions during an online class, the AITA can respond quickly, creating an interactive dialogue that keeps the student engaged. Similarly, for intuitives, the AITA could provide discussion prompts or simulate peer discussions to explore abstract ideas in depth. Moreover, the AITA can help thinkers refine their learning strategies by providing feedback and performance analysis. Perceivers, meanwhile, will benefit from the AITA's ability to remind them of upcoming deadlines and assist in organising their study schedules.

Second, online platforms can enhance student interactions by incorporating grouping functions. As proven by previous studies, group collaborative learning within traditional classrooms can help students understand knowledge and promote learning (Miller & Hadwin, 2015). For this reason, we advocate for online education platforms to adopt a similar approach by allowing the formation of student groups—whether through system-generated randomness, student self-organisation, or teacher assignments. Upon a teacher posing questions or tasks in class, students can join groups and exchange their views directly on the platform. To further enrich this function, the introduction of an "ask for help" button is advised, which allows group members to call the teacher for guidance. During the discussion, students can voice their opinions, engage critically with diverse viewpoints, and access timely support when needed, thus enhancing the engagement levels and quality of the online learning experience.

Third, some tools can also be added to online education platforms to ensure timely feedback between teachers and students. For example, a "Do not understand" button could be set under each slide. Students can click this button when they encounter difficulties in understanding the content, and it instantly provides feedback to the teacher's device. In addition, some platforms already allow students to send real-time Danmaku comments, allowing teachers to master students' comprehension (Chen et al., 2017). However, an excessive number of Danmaku comments can cause visual distraction and cognitive workload. We propose that the feedback efficiency could be improved by integrating a word cloud analysis function that can extract keywords from many comments. We also recommend that teachers use the platform's virtual currency, which cannot be withdrawn but can be exchanged for other learning resources, as an incentive to encourage active class participation among students.

4.3 Limitations and future directions

Although the current study enhances our understanding of learning performance among different personality types of students who are studying online, it has certain limitations. First, the participants in this study were limited to first-year high school students in Hainan Province, China. Students from diverse regions and cultural backgrounds, as well as students of different ages, may exhibit varying online

learning challenges and needs. Hence, it is advisable to approach generalizing the conclusions with caution. It would be beneficial for future research to incorporate a wider range of ages and cultural backgrounds to explore whether these factors interact with personality type in influencing online learning performance.

In addition, future studies can explore how students' socioeconomic background (e.g., parental education level, family income) moderates the impact of personality type on online learning performance. Existing research indicates that students' social class affects the learning environment and conditions they encounter while taking courses at home (Goudeau et al., 2021; Haelermans et al., 2022). Students from lower socioeconomic backgrounds may be more likely to fall behind or face challenges due to factors such as the inability to afford reliable broadband connection, the responsibility of doing household chores, or the lack of sufficient study space (Hoskins & Wainwright, 2023). Furthermore, compared to working-class parents, upper- or middle-class parents tend to provide higher levels of academic support, including organising educational activities designed to develop their children's learning skills and spending more time helping with homework (Golden et al., 2023). As online learning heavily relies on the family, incorporating additional socioeconomic background information could enable more precise prediction of individual online learning performance based on a student's personality type.

Another limitation of this study is that it only provides short-term performance evaluations, while learning performance may evolve over time. Research has shown that students may increasingly adapted to online learning by adopting new learning strategies tailored to the online environment and becoming more proficient in using digital tools (Mou, 2023). In parallel, educational institutions have implemented a range of measures (e.g., training teachers in delivering effective online instruction) to improve student engagement and the quality of online education (Srivastava et al., 2024; Sato et al., 2024). Thus, long-term follow-ups are recommended to provide a more comprehensive understanding of how personality type influences online learning performance over an extended period.

5 Conclusion

This study employs large-scale empirical data to investigate the influence of students' personality types on their online learning performance. The obtained findings indicate that the online learning environment confers disadvantages to extroverts, intuitives, thinkers, and perceivers. The characteristics of a subject determine the direction and extent of the impacts of personality type on online learning performance in that subject. Several significant interaction effects between different MBTI dimensions on online learning performance were revealed. No sex differences were found in the effect of personality type on online learning performance, yet girls outperformed boys in online learning. The current study provides a valuable reference for identifying students who may encounter more obstacles within the context of online learning so that we can pay more attention to and support them. It also provides guidelines for designing effective tools for online education platforms to improve their learning performance.

Appendix

Appendix Section A: Tables

Table 5 Appendix Table A.1. Seven level-1 high schools in Hainan Province

School name	Location	Total number of students	Number of students completing the questionnaires	Number of students without exam results	Number of students with problems in MBTI responses	Number of valid participant
High School Affiliated to Hainan Normal University	Haikou	738	641	8	0	633
Huangliu High School Affiliated to East China Normal University	Ledong	771	771	3	13	755
Lingshui High School Affiliated to Minzu University of China	Lingshui	310	298	4	0	294
Hainan Middle School Sanya Campus	Sanya	335	318	13	0	305
Wanning Middle School Affiliated to Beijing Normal University	Wanning	730	727	1	14	712
Wenchang Middle school	Wenchang	1139	1106	96	18	992
Yangpu Middle school	Danzhou	668	663	8	6	649

Table 6 Appendix Table A.2. Personality types in the four dimensions among students of different sex

		Boys		Girls		*X-squared*	*p*
		N	Proportion	*N*	Proportion		
EI	E	1091	55.16%	1226	51.91%	0.102	0.750
	I	887	44.84%	1136	48.09%		
SN	S	1273	64.36%	1711	72.44%	1.160	0.282
	N	705	35.64%	651	27.56%		
TF	T	926	46.81%	952	40.30%	0.617	0.432
	F	1052	53.19%	1410	59.70%		
JP	J	923	46.66%	1148	48.60%	0.018	0.894
	P	1055	53.34%	1214	51.40%		

The proportion refers to the percentage of students identified with a certain personality type within each sex

Table 7 Appendix Table A.3. Sixteen personalities of students of different sex

	Boys		Girls		Total	
	N	Proportion	*N*	Proportion	*N*	Proportion
ENFJ	51	2.58%	73	3.09%	124	2.86%
ENFP	194	9.81%	166	7.03%	360	8.29%
ENTJ	82	4.15%	67	2.84%	149	3.43%
ENTP	98	4.95%	77	3.26%	175	4.03%
ESFJ	149	7.53%	224	9.48%	373	8.59%
ESFP	172	8.70%	263	11.13%	435	10.02%
ESTJ	281	14.21%	247	10.46%	528	12.17%
ESTP	64	3.24%	109	4.61%	173	3.99%
INFJ	34	1.72%	43	1.82%	77	1.77%
INFP	115	5.81%	125	5.29%	240	5.53%
INTJ	46	2.33%	34	1.44%	80	1.84%
INTP	85	4.30%	66	2.79%	151	3.48%
ISFJ	139	7.03%	228	9.65%	367	8.46%
ISFP	198	10.01%	288	12.19%	486	11.20%
ISTJ	141	7.13%	232	9.82%	373	8.59%
ISTP	129	6.52%	120	5.08%	249	5.74%

The proportion refers to the percentage of students identified with a certain personality type within each sex/total

Appendix Section B: The MBTI Questionnaire for High School Students

Instructions: Please read each question carefully and put a checkmark on the corresponding option letter. There are no right or wrong options, so do not overthink the question and answer it based on your first impression.

1. In the circles of classmates and friends, you _____?
   A. Usually, finding out about others' news quickly
   B. Gossip is slow to catch up
2. You are more inclined to choose _____?
   A. To maintain the objective
   B. To be sympathetic
3. Which writing style do you prefer?
   A. Unique and innovative expression
   B. Direct and clear expression
4. When facing others, you tend to be _____?
   A. Kind to people
   B. Fair to people
5. When talking with friends, you usually?
   A. Initiate the conversation
   B. Waiting for others to start
6. When making decisions, you are more concerned with _____?
   A. Whether it is objective and fair
   B. Do you and others feel good about it
7. You prefer _____?
   A. To think and do things on the fly
   B. To plan tasks step by step in advance
8. Meeting new faces is more likely to make you _____?
   A. Feel pressured and burdened
   B. Feel excited and interested
9. In a group, you usually _____?
   A. Interact only with those you know well
   B. Interact with many people, including strangers
10. You prefer things to be _____?
   A. Neat and orderly
   B. Casual and free
11. Which are you more interested in?
   A. Abstract concepts
   B. Concrete examples
12. Which happens more often to you?
   A. Emotions dominate logic
   B. Logic-dominated emotions

13. You prefer _____?
   A. Live environments
   B. Quiet environments
14. Which word fits you better?
   A. Practical
   B. Ingenious
15. When meeting strangers for the first time, you usually _____?
   A. Do not know what to say
   B. Can chat quite easily
16. In most cases, you are a person who _____?
   A. Likes to daydream
   B. Is practical
17. You are generally a person who is _____?
   A. Emotionally intense
   B. Clear-minded
18. You like to spend more time _____?
   A. With others
   B. Alone
19. Which attracts you more?
   A. Explicit expressions
   B. Implications
20. You prefer _____?
   A. Detailed planning
   B. Going with the flow
21. When others show enthusiasm towards you upon first meeting, you usually _____?
   A. Respond reluctantly
   B. Respond enthusiastically
22. You prefer _____?
   A. Being imaginative
   B. Sticking to the facts
23. Friends consider you more inclined to be _____?
   A. Understanding
   B. Analytical
24. Which is more like you?
   A. Usually, plan ahead to avoid pressure
   B. Enjoying emergencies that require immediate attention
25. Which description fits you better?
   A. Practical
   B. Innovative
26. Which description fits you better?
   A. Can open up to most people
   B. Are reserved for most people

27. If a classmate suddenly calls you to hang out on the weekend, you would _____?
   A. Check your schedule; cannot disrupt your plans
   B. Be up for anything
28. Following a daily schedule _____?
   A. Suits you well
   B. Feels constraining
29. When meeting a group of friends for the first time, you are more likely to _____?
   A. Wait for someone else to introduce you
   B. Proactively introduce yourself
30. Which description fits you better?
   A. Like-seeking novelty and uniqueness
   B. Prefer sticking with what is familiar
31. When making decisions, you value _____ more.
   A. Principles
   B. Personal feelings
32. Faced with a must-do task, you tend to _____.
   A. Do it first, then play
   B. Play first and do it at the last minute
33. When choosing student clubs, you care more about whether _____.
   A. The members are friendly and nice
   B. The members are capable
34. Which description fits you better?
   A. Your desk is tidy and organised
   B. Your desk is cluttered and disorganised
35. You feel you are more _____.
   A. Rational
   B. Emotional
36. Most of the time, you are _____.
   A. Relaxed and casual
   B. Orderly and planned
37. When you go to meet strangers, you usually feel _____.
   A. Somewhat excited
   B. Reluctant
38. You are more inclined to choose something _____.
   A. Real
   B. Imaging
39. Which is more like you?
   A. You care a lot about others' feelings and thoughts
   B. You focus on solving the problem
40. In a group, you tend to _____.
   A. Be the centre of attention
   B. Keep to yourself and do your own thing

41. If you have free time, you prefer _____.
   A. Attending various gatherings or club activities
   B. Spending time alone enjoying personal time
42. When facing a problem, you prefer _____.
   A. Looking for new, creative solutions
   B. Relying on practical solutions based on past experience
43. You prefer people who are _____.
   A. Logical
   B. Emotionally rich
44. When tasked with an assignment, you often _____.
   A. Start preparing early
   B. Wait until it is necessary to start
45. You think it is more important to _____.
   A. Be realistic
   B. Anticipate the future
46. Which description fits you better?
   A. Enjoy making plans
   B. Act spontaneously
47. During group activities, you tend to _____.
   A. Mingle with everyone
   B. Stay quiet
48. When admiring flowers, you are more likely to _____.
   A. Focus on their colour, shape, and scent
   B. Get lost in the imagination or feelings they evoke
49. When organising large events, your greatest strength is _____.
   A. Making detailed plans
   B. Adapting flexibly to changes
50. In a team needing production and design, you'd rather participate in _____.
   A. Production
   B. Design
51. Which word do you prefer?
   A. Touching
   B. Convincing
52. Regarding holiday homework, you usually _____.
   A. Start to feel motivated to do so as the holiday ends
   B. Plan ahead and do it on schedule
53. Classmates often describe you as _____.
   A. Lively and energetic
   B. Quiet and reserved
54. You are more attracted to stories that are _____.
   A. Logically sound
   B. Full of emotions
55. During vacations, you hope to _____.

A. Have gatherings and trips planned out early
B. Be free and spontaneous, doing whatever seems fun at the moment
56. Under normal circumstances, you more easily notice _____.
A. The issues at hand
B. Future possibilities
57. When chatting with friends, you spend more time _____.
A. Listening
B. Talking
58. In discussions, you usually _____.
A. Have many ideas and divergent thoughts
B. Focus closely on the issue being discussed
59. What usually persuades you more?
A. A logically coherent argument
B. A heartfelt plea
60. Which state do you enjoy more?
A. Letting things happen naturally
B. Everything going according to plan
61. You're more willing to make friends with people who are _____.
A. Sharp-tongued but always makes sense
B. Gentle but often lack logic
62. Which description fits you better?
A. Set a goal and then do your best to achieve it
B. Frequently change goals, liking new experiences
63. You tend to choose solutions based on _____.
A. Past experiences
B. New ideas regardless of past experiences
64. What is more likely to happen to you?
A. Overly critical
B. Overly emotional status
65. Most of the time, you are always _____.
A. Focused on the present
B. Considering various future possibilities
66. Which type of friend attracts you more?
A. Those who can engage in intellectual discussions and debates
B. Those who can understand and empathise with emotions
67. Which state do you prefer?
A. Organised
B. Spontaneous
68. Most of the time, you tend to _____?
A. Avoid showing emotions
B. Communicate your emotions to others anytime
69. In class, you usually _____?
A. Actively participate in various group activities
B. Only participate in activities that interest you

70. When a friend is in trouble, what kind of support do you mainly provide?
   A. Emotional support
   B. Analytical and solution-oriented support
71. In the eyes of others, you are always more _____?
   A. Rushed than leisurely
   B. Leisurely than rushed
72. Compared to the other, you would rather believe in _____?
   A. Things that can be touched and seen
   B. Intuition and inspiration
73. You prefer to be friends with people who are _____?
   A. Down-to-Earth
   B. Creative
74. Which description fits you better?
   A. Talkative
   B. Silent
75. Which description fits you better?
   A. Organised
   B. Unconcerned with minor details
76. In which situation do you tend to notice first?
   A. When a classmate's expression is illogical
   B. When a classmate needs emotional support
77. Overall, you see yourself as someone who _____?
   A. Is more focused on the external world, energised by interacting with people?
   B. Is more focused on the inner world, energised by spending time alone
78. Overall, you see yourself as someone who _____?
   A. Pays more attention to factual details and practicality
   B. Values the underlying meanings and possibilities of things
79. Overall, you consider yourself _____?
   A. Someone who makes decisions based on facts, rules, and logic, caring about fairness and reason
   B. Someone who makes decisions based on emotions, feelings, and values, valuing humaneness
80. Overall, you view yourself as _____?
   A. Someone who likes to plan ahead, making life more organised
   B. Someone who enjoys flexibility, frequently changing plans to keep life adaptable

Author contributions Xin Wu and Zhenhua Chen led the data collection and management. Xin Wu contributed to the questionnaire design. Yuzhen Wang, Yonghao Huang and Ruifeng Yu contributed to the statistical analysis. Yuzhen Wang drafted the manuscript. Ruifeng Yu and Yuzhen Wang critically revised the manuscript. All the authors were responsible for the decision to submit this manuscript for publication.

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Data Availability The dataset that supports this study cannot be made openly available because it contains students' personal information. Interested researchers may contact the corresponding author to request access.

Declarations

Compliance with Ethical Standards This study was approved by the Institutional Review Board of Tsinghua University and complied with all relevant ethical regulations.

Competing interests The authors have no competing interests to declare that are relevant to the content of this article.

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